

Midterm 1

ORF 309, Spring 2025

Instructions: This midterm has 3 questions for a total of 90 points. Partial credit will be awarded for partial proofs or arguments. Strive to reduce answers as much as possible (e.g., give a fraction in simplest terms, though a numerical answer requiring a calculator is not necessary). You may freely use basic facts from class, such as laws of probability, properties of conditional expectation, or manipulations with pmfs/pdfs. However, you must provide adequate justifications for important steps. Put another way, try your best to demonstrate awareness that you are applying the correct facts in the correct way by explicitly mentioning the results you are applying, indicating the objects they are applied to, and checking the required assumptions.

Length: 75 minutes (for working on the problems) + 15 minutes (for submission)

Good Luck!

Question	Points	Score
1	30	
2	30	
3	30	
Total:	90	

1. Bob is going to take an exam for his Probability course. There are 2 questions. Bob's performances on the two questions are not mutually independent, as, clearly, both depend on his understanding of the course materials. Bob estimates that he has probability 0.8 to answer the first question correctly. Given he answers the first question correctly, he has probability 0.6 to answer the second question correctly. If he gets the first question wrong, then he has probability only 0.1 to answer the second question correctly.
 - (a) (10 points) What is the probability that Bob answers the second question correctly?
 - (b) (10 points) Suppose we know that Bob answers the second question correctly. Given this information, what is the probability that he answers the first question correctly?
 - (c) (10 points) Each question is worth 10 points. Bob will get the points if he answers a question correctly, and no point otherwise. What is the variance of Bob's grade?

2. Trains at Princeton Junction Station often experience unpredictable delays. Suppose that the delay times (in minutes) of the first scheduled trains in the afternoon to New York and Trenton are both geometrically distributed, with parameters $p = 0.1$ and $q = 0.2$, respectively (as the clock at the train station only shows minutes, not seconds, we model the delay times as discrete random variables). The delays of the two trains are probabilistically independent.
- (a) (10 points) Suppose that we know the train to New York is delayed by a time strictly longer than 10 minutes. Given this information, what is the probability that the total delay time of that train is strictly longer than 20 minutes?
 - (b) (10 points) Suppose we know that the train to Trenton is delayed by 5 minutes. Given this information, how long do we expect the train to New York to be delayed?
 - (c) (10 points) Calculate the probability that the train to New York is delayed by a strictly shorter time than the train to Trenton.

3. Alice invests in Fruity Yogurt Inc. by holding several shares of its stock and cares about the evolution of the stock price. The stock price of Fruity Yogurt Inc. follows a mysteriously discrete pattern. On each day, the stock price only changes once, **exactly at noon**; the stock price will be either doubled (with probability 0.4) or halved (with probability 0.6). Suppose that the price movements of different days are mutually independent. **After** the noon of today (which represents day 0), Fruity Yogurt's stock price is \$1.
- (a) (10 points) Calculate the probability that **after** the noon of day 5, the stock's price is \$2. (Hint: think about how many days out of the 5 days should the price go up?)
 - (b) (10 points) If n is very large number, estimate the fraction of trading days (among the first n trading days) on which the stock price is doubled.
 - (c) (10 points) Calculate the expectation of the stock price **after** the noon of day n . What happens as $n \rightarrow \infty$?